Optimal Design of Flexible & Reliable Process Systems

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Key Words — **Process design, Process flexibility, Batch plant; Process optimization**

Summary & Conclusions — **In design & operation of chemical process systems, two categories of uncertainty are important: 1) realization of continuous process parameters (product demands, feedstock qualities, reaction kinetics), and 2) discrete states (typically related to equipment availability). This paper presents a general framework for incorporating both types of uncertainty in process system design. A combined flexibility-reliability index (FRI) is introduced to measure the capability of a process system to cope simultaneously with 1) (stochastic) parameter variations, and 2) equipment failures, by explicitly accounting for process operational interactions & constraints. A design optimization procedure is then proposed for obtaining flexible, reliable process systems at minimum investment cost and maximum average profit; the potential inclusion of maintenance considerations is briefly discussed.**

1. INTRODUCTION

Traditional reliability assessment of process systems provides quantitative information on the probability of system failures based on the reliability characteristics of individual elements and on a system model. Availability assessment addresses the fraction of system uptime in a given time interval by accounting for maintenance aspects. System effectiveness involves the assessment of the average production capability by including unit production capacities [1]. In existing techniques, the production capacity is typically defined for a single product in direct relation to equipment availability & capacity [2]. For complex multiproduct process systems, however, two major questions arise:

- how to measure system effectiveness by explicitly accounting for potential interactions between subsystems, in the presence of continuous parameter uncertainty,
- how to obtain an optimal design which features an optimal/desired level of system effectiveness.

Due to continuous market pressures, process environments (*eg*, chemical & refinery plants or pharmaceutical & food processes) are typically faced with continuous parameter uncertainty corresponding to variations in:

- external parameters (*eg*, feed-stream quality, product demands, environmental conditions, prices),
- internal process parameters (*eg*, heat-transfer coefficients, reaction constants, physical properties, processing times).

These variations are usually specified through fixed ranges or continuous probability distributions. The ability of a process system to accommodate continuous uncertainty is *process flexibility* [3, 4]. Process flexibility accounts for process interactions and can readily be used in complex multiproduct environments. However, it does not explicitly consider reliability & maintenance aspects.

The need for coupling flexibility with reliability has only recently been realized in various disciplines, such as,

- production manufacturing [5],
- product quality management [6],
- chemical engineering [7, 8].

This need calls for the development of a systematic integrated framework for measuring flexibility, reliability, and availability. For such a measure to be meaningful & useful in practical applications, it must be based on:

- a thorough mathematical model of the process;
- a description of the uncertainty involved, both continuous & discrete, *eg*, equipment availability;
- an explicit consideration of service rates and maintenance policies.

This paper presents the initial development of an optimization-based approach for designing process-plants that are flexible & reliable to cope with continuous variations and equipment failures. Section 2 presents a motivating example, the design of a multiproduct batch plant (widely used in the pharmaceutical & food industries), which: 1) introduces the need for such an integrated framework, and 2) reveals some complex features of the operation of such plants in the presence of continuous & discrete uncertainties. Section 3 presents the concepts of process flexibility and the $\mathcal{FRI}$. An extension of the concepts to include maintenance aspects are briefly discussed. Section 4 proposes a formal design optimization to obtain a flexible, reliable design at minimum investment cost or maximum average profit.

Acronyms

$\mathcal{FOR}$	feasible operating region (compact set of feasible points of the system)
$\mathcal{FRI}$	combined flexibility-reliability index
$\mathcal{SFI}$	stochastic flexibility index.

Notation

A_i, B_i	constant cost-coefficients
$C_I(d)$	annualized investment cost function
FR_H	$\mathcal{FRI}$ over $\mathcal{H}$
$\mathcal{H}$	time horizon of interest (all products are processed)
I	set of desired products, $i \in I$
J	set of stages of the batch plant, $j \in J$
$J(\Theta)$	joint pdf of the continuous uncertain-parameters, Θ

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K	set of potential discrete operating states
K^{OP}	set of operable operating states, $K^{OP} \subset K$
$P_k(t)$	state probabilities as a function of time
P_k^{ss}	steady-state probabilities
Q_i	demand for product i
$\mathfrak{R}$	s-expected revenue
SF_k	$\mathcal{SFI}$ of state k
$SF(t)$	time-dependent $\mathcal{SFI}$
$S_{i,j}$	size factor; amount of product i required in stage j
T_{L_i}	limiting cycle-time of product i; see (1a)
V_j	equipment volume
$\boldsymbol{d}$	vector of design variables (amounts of equipment)
m_j	maximum number of units in stage j
n_j	operable pieces of equipment in stage j, $0 \le n_j \le m_j$
$r_j(t)$, $q_j(t)$	[reliability, unreliability] of equipment in stage j
$t_{i,j}$	processing time of product i in stage j
$\mathcal{B}_i$	batch size of product i
$\mathcal{S}_k$, $\bar{\mathcal{S}}_k$	index set of [operable, non-operable] parts of the equipment in state k
Θ	set of continuous uncertain parameters
$\Phi(h)$	pdf for a single r.v., h.

Other, standard notation is given in "Information for Readers & Authors" at the rear of each issue.

MOTIVATING EXAMPLE

A Multiproduct Batch Plant

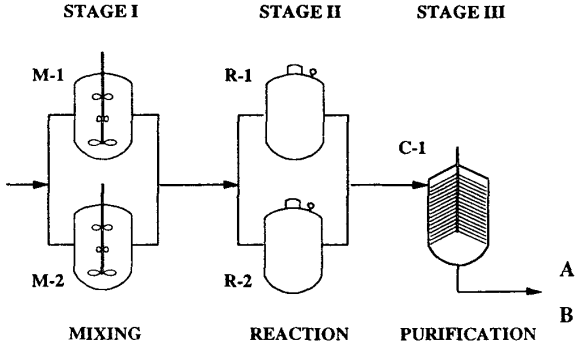


Figure 1. Multiproduct Batch Plant

2.1 General Description

Consider the following multiproduct batch plant, as shown in figure 1. It involves 3 main processing stages (mixing, reaction, purification) for manufacturing 2 different products (A, B).

- Stage #1, mixing: 2 mixers (M-1, M-2) in *parallel*[1] mix the input raw materials;
- Stage #2, reaction: 2 reactors (R-1, R-2) in *parallel*, convert the raw materials to intermediate products;
- Stage #3, purification: a centrifuge (C-1) purifies the intermediate products to final products with the desired specifications. ◄

Such an arrangement is common in the pharmaceutical, food, biochemical, and specialty chemical industries, where several different products are manufactured by using the same parts of equipment at different time instances [9].

Although the structural arrangements of a batch plant are rather simple (especially from a reliability point of view, one would be tempted to argue that it is a simple *series-parallel* configuration), its operation is far from trivial because process-interactions play a key role in the proper assessment of system behavior. This is mainly attributed to:

- Each product is processed in batches sequentially following each stage; for the following product to be processed it has to wait until the processing of previous product is over and the units are cleaned.
- The time to process a batch of product on a unit of a processing stage is *processing time*. If more than one piece of equipment is available per stage, the processing time reduces according to the number of equipments (if all are used).
- The mean time between the processing of two successive batches of the same product is *limiting cycle time*; this time controls the total production time (for each product and overall) within a given horizon time $\mathcal{TC}$. ◄

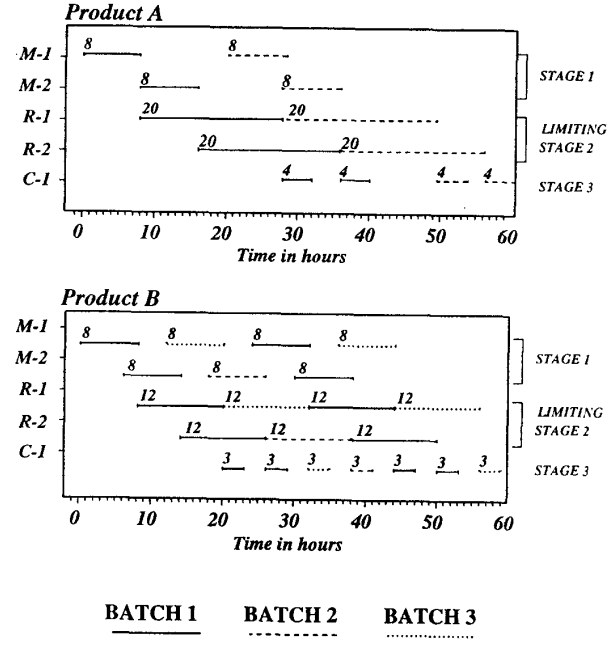


Figure 2. Gantt Chart for Products A, B

The operation scheduling of such a multiproduct batch plant is typically represented by a Gantt chart as shown in figure 2,

[1]The terms, *series* & *parallel* are used in their logic-diagram sense, irrespective of the schematic-diagram or physical-layout.

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TABLE 1
Data for Example (initial design)

Stage	Pieces of Equipment	Size (liters)
Mixing	2	800
Reaction	2	1200
Purification	1	1600

Limiting Cycle Time (hours)		
Operating State	Product A	Product B
2-2-1	10	6
1-2-1	10	8
2-1-1	20	12
1-1-1	20	12
Batch size (kg)	400	200

TABLE 2
Processing Times and Size Factors for Example Problem
[$\mathcal{H}$ = 6k hours]

	Processing Time (hours) Stage			Size factor (liters/*tone*) Stage		
Product	1	2	3	1	2	3
A	8	20	4	2000	3000	4000
B	8	12	3	4000	6000	3000

for the design data of tables 1 & 2. The *limiting cycle time* for product A & B are 10 hours & 6 hours respectively (the reaction stage is the limiting step). For example, the process for product A is:

- a batch is produced (processed) in 8 hours in the first mixer M-1,
- it enters the reaction stage (reactor R-1), and remains for 20 hours,
- it enters the purification section, and remains for 4 hours.

Since there is only 1 centrifuge, the utilization of M-2 (mixer #2) and R-2 (reactor #2) is delayed, yet the reaction time poses the limit in operation: the processing steps cannot proceed any faster until the reaction is completed.

Now, suppose that M-2 is not available, state (1-2-1); then, the operation of the batch plant *changes*, as depicted in figure 5 with limiting cycle times of 10 & 8 hours for products A & B respectively (stage 1 is now the limiting stage for product B). These new scheduling requirements in the production operation of the plant are essential to assess properly the performance of the multiproduct batch plant subject to equipment (un)availability. Furthermore, if product demands can only be forecast (with the data shown in table 3) then the evaluation of the process performance of the batch plant of figure 1, in satisfying both the uncertain demand and equipment failures, requires simultaneous consideration of process flexibility, batch scheduling, and reliability characteristics.

TABLE 3
Demand & Cost Data for Example

	Demand (10^3 kg/$\mathcal{H}$)		
Product	μ	σ	Price $/$10^3$ kg
A	40.0	10.0	5000
B	60.0	15.0	4000

2.2 Mathematical Model

Assumptions

1. Intermediate storage tanks are not available.
2. Effects of transfer times can be neglected.
3. All products follow the same sequence of processing stages.
4. Eq (1) governs the system. See also [9].

$$T_{L_i} = \max_{j\in J}\{t_{i,j}/n_{i,j}\}, \text{ for } i \in I \tag{1a}$$

$$\mathcal{B}_i = \min_{j\in J}\{V_j/S_{i,j}\}, \text{ for } i \in I \tag{1b}$$

$$\sum_{i\in I} Q_i \cdot T_{L_i}/\mathcal{B}_i \le \mathcal{H}, \tag{1c}$$

$$C_I = \sum_{j\in J} A_j \cdot m_j \cdot V_j^{B_j}, \tag{1d}$$

$$\mathcal{R} = \sum_{i\in I} p_i \cdot Q_i. \tag{1e}$$

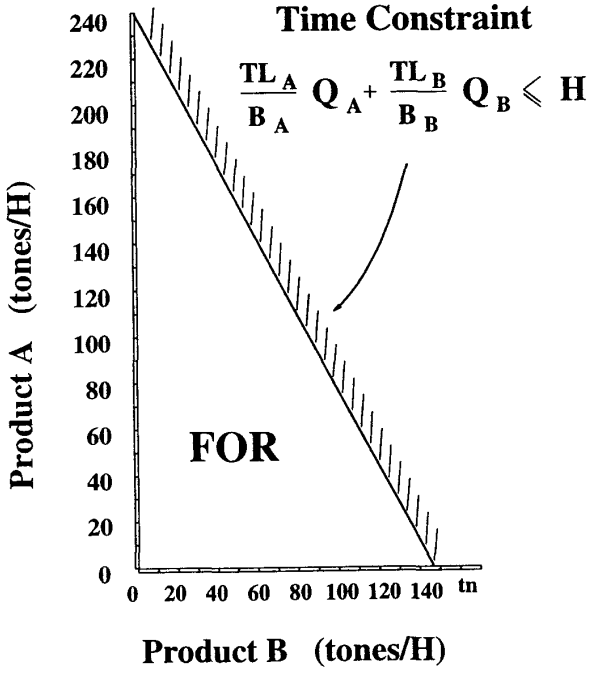
Constraint (1a) approximates the limiting cycle time to the largest processing time (divided by the number of parallel units). Constraint (1b) defines the batch size of product i. The time horizon constraint (1c) relates the maximum production time of each product i to the overall available time $\mathcal{H}$. Constraints (1d) & (1e) represent the cost & revenue functions, respectively. ◄

An interesting feature of (1) is that the $\mathcal{FOR}$ in the space of the product demand Q_i is linear (for fixed V_j, m_j) with the horizon time constraint (1c) limiting the plant capability. This is illustrated in figure 3 (for the example problem of figure 1 with the data of table 1); in this case, the batch plant can satisfy a range of demand realizations (240 10^3 kg of product A, 140 10^3 kg of product B, or a mix of both). This feature is used to advantage in the flexibility-reliability analysis, which is presented in the next section.

3. SIMULTANEOUS FLEXIBILITY-RELIABILITY ANALYSIS

For continuous uncertain parameters (uncertain product demands), typically approximated by a probability distribution, the question is how to evaluate the capability of a process system

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FOR: Feasible operating region in the space of product demands

Figure 3. $\mathfrak{FOR}$ for the Multiproduct Batch Plant

to cope effectively with their random realizations. In general, this requires determination of the intersection between the plant $\mathfrak{FOR}$ and the joint pdf $J(\Theta)$. The $\mathfrak{SFI}$ [4] precisely corresponds to this estimation:

$$\mathrm{SF} = \int_{\Theta \in \mathfrak{FOR}} J(\Theta)\, d\Theta. \tag{2}$$

In determining SF, the identification of the $\mathfrak{FOR}$ of the process design is required. This itself is a complex task involving the solution of optimization problems [4]; for a multiproduct batch-plant design model with uncertain product demand, however (due to the linear time horizon constraint) the evaluation of the $\mathfrak{SFI}$ reduces to [7]:

$$\mathrm{SF} = \int_{-\infty}^{\mathfrak{IC}} \Phi(h)\, dh, \tag{3}$$

$$h \equiv \Sigma_i\, Q_i \cdot T_{L_i} / \mathfrak{B}_i,$$

the mean & variance/covariance are defined through the linear transformations.

Such a $\mathfrak{SFI}$ represents the maximum probability of a design to meet process constraints & specifications. In this respect, $\mathfrak{SFI}$ also represents the reliability of the process system at a fixed system (equipment) state in the presence of continuous uncertainty.

Since any equipment failure defines a different system state with an associated $\mathfrak{SFI}$, then by modeling the random life-length of equipment, the flexibility as a function of time can be established. An $\mathfrak{FRI}$ can then be introduced which measures the average capability of the process design to accommodate simultaneously the uncertainty in the equipment availability and the process parameters. An algorithm for determining such a $\mathfrak{FRI}$ is summarized in the following basic steps.

Algorithm

Step 1. Define all potential operating states of the system with an associated $\mathfrak{SFI}$,

$$\mathrm{SF}_k \geq 0,\ k \in K^{\mathrm{OP}}.$$

Step 2. Estimate the (un)reliability of each part of the equipment as a function of time,

$r_j(t)$, $q_j(t)$. For example,

$$\mathrm{r_j(t)} = \int_t^{\infty} \mathrm{weif}(t/\alpha_j;\ \beta_j)\, dt,\ j \in \mathfrak{S}_k,$$

$$q_j(t) = \int_0^{t} \mathrm{weif}(t/\alpha_j;\ \beta_j)\, dt,\ j \in \bar{\mathfrak{S}}_k.$$

Step 3. Determine the probability of each system state k, *eg*, assuming *s*-independent equipment failures:

$$P_k(t) = \prod_{j \in \mathfrak{S}_k} r_j(t) \cdot \prod_{j \in \bar{\mathfrak{S}}_k} q_j(t),\ k \in K.$$

For multiproduct batch plants,

$$P_k(t) = \prod_{j \in J} \binom{m_j}{n_j^k} \cdot r_j(t)^{n_j^k} \cdot q_j(t)^{m_j - n_j^k},\ k \in K.$$

Step 4. Calculate $\mathrm{SF}(t)$:

$$\mathrm{SF}(t) = \sum_{k \in K^{\mathrm{OP}}} P_k(t) \cdot \mathrm{SF}_k.$$

Step 5. Determine $\mathrm{FR}_{\mathfrak{IC}}$, $\mathfrak{IC} = [T_a, T_b]$:

$$\mathrm{FR}_{\mathfrak{IC}} = \frac{1}{\mathfrak{IC}} \cdot \int_{T_a}^{T_b} \mathrm{SF}(t)\, dt$$

$$= \frac{1}{\mathfrak{IC}} \cdot \int_{T_a}^{T_b} \sum_{k \in K^{\mathrm{OP}}} P_k(t) \cdot \mathrm{SF}_k\, dt.$$

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Qualitatively, $\mathcal{FRJ}$ represents the average flexibility of the process design over all possible system states within the specified $\mathcal{HC}$. Therefore, it measures the average system performance under both continuous uncertainty and random equipment failures. The closer this $\mathcal{FRJ}$ is to the $\mathcal{SFJ}$ at the initial (no equipment failure) state, the more reliable the system is. From the current expression for $FR_{\mathcal{HC}}$, as $\mathcal{HC} \to \infty$, the $\mathcal{FRJ} \to 0$. This is consistent with the assumption of no-repair-mechanism for the equipment. In this respect, the $\mathcal{FRJ}$ corresponds to a worst-case situation. Incorporation of maintenance is a straightforward extension within this framework, and is briefly discussed next.

3.1 Incorporation of Maintenance

Taking maintenance aspects into account, the steps of the $\mathcal{FRJ}$ determination need slight modification. In particular, in a continuous operating environment where failed equipment is maintained, a steady-state system model can be used instead of the time-dependent reliability functions in step 3 of the Algorithm. Thus, the P_k^{ss}, for each system state must be determined, *eg*, through a Markov analysis of system states, for different maintenance scenarios (assuming for example constant failure & repair rates) by solving the set of corresponding linear frequency balance equations [10]. A $\mathcal{FRJ}$ can then be defined as:

$$FA = \sum_{k \in K} P_k^{ss} \cdot SF_k. \tag{4}$$

As the repair policy affects the balance equations (and thus the values of P_k^{ss}), the $\mathcal{SFJ}$ is a means of comparing repair policies. Aggregation techniques can also be applied to reduce the large number of possible system states, especially for the incorporation of more realistic reliability & maintenance models [11]. Maintenance optimization aspects are discussed elsewhere [12].

3.2 Multiproduct Batch Plant Example

The application of these tools to the multiproduct batch plant design of figure 1 is considered here. By assuming variations of the product demand following s-normal distributions (see table 2), the evaluation of $\mathcal{SFJ}$ gives SF=0.9946 (integrating over the region $\mu \pm 4\sigma$); this implies that the design can accommodate 99.46% of the s-expected product demand variations, figure 4.

TABLE 4
Cost & Reliability Data for Example Problem

	parameters for $C_I = A_j \cdot V_j^{B_j}$		parameters for weif(u/α; β)	
j	A	B	α	β
1	250	0.6	65000	0.50
2	450	0.6	80000	0.60
3	350	0.6	110000	0.80

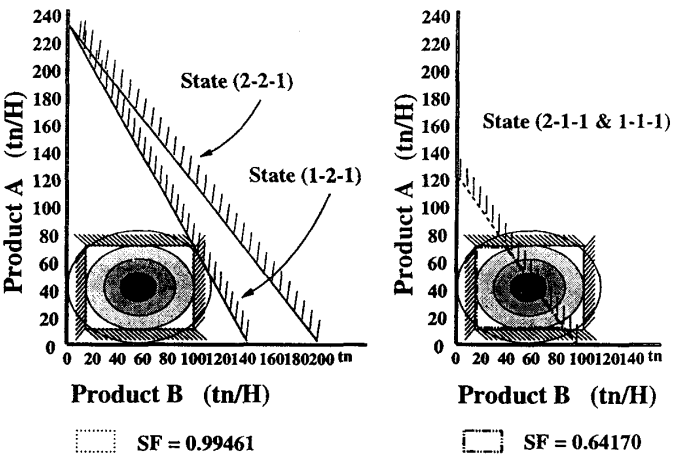


Figure 4. Effect of Equipment Failures on $\mathcal{FOR}$

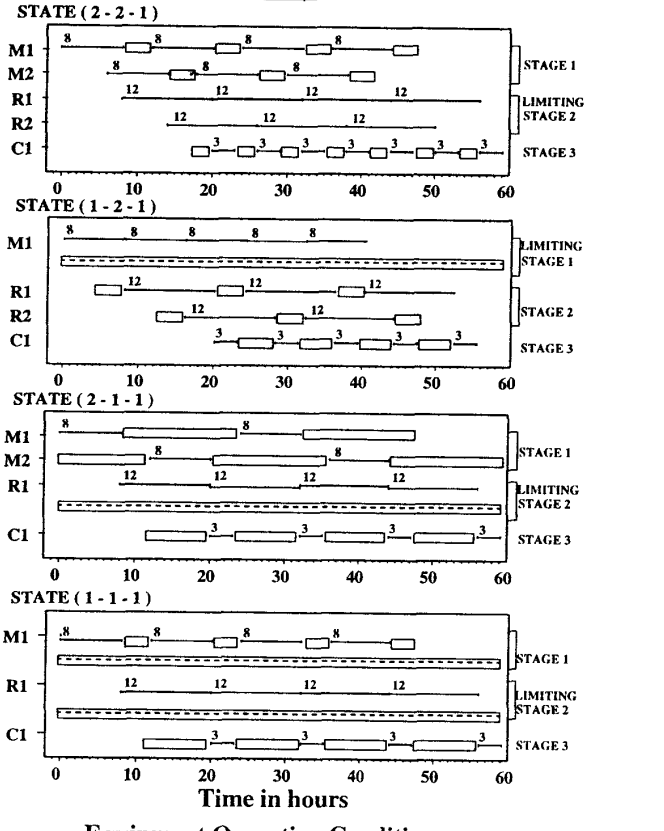
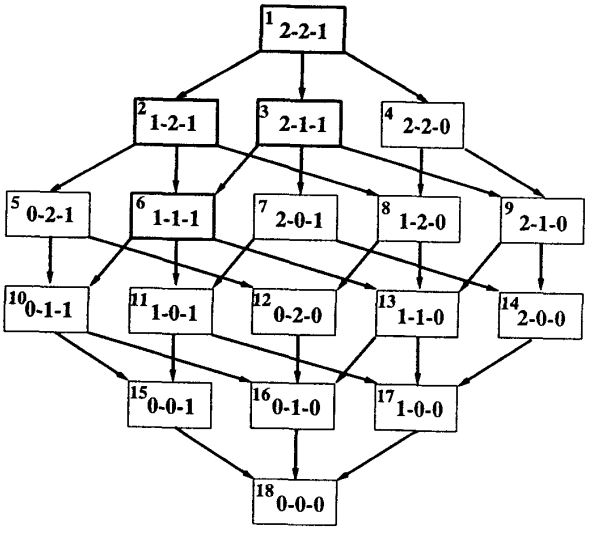


Figure 5. Effect of Equipment Unavailability on Batch Operation (Product B)

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State	Aggregation	Operable	State	Aggregation	Operable
1	1	yes	10	2	-
2	2	yes	11	2	-
3	2	yes	12	1	-
4	1	-	13	4	-
5	1	-	14	1	-
6	4	yes	15	1	-
7	1	-	16	2	-
8	2	-	17	2	-
9	2	-	18	1	-

Figure 6. State Network and Transitions Due to Equipment Failure

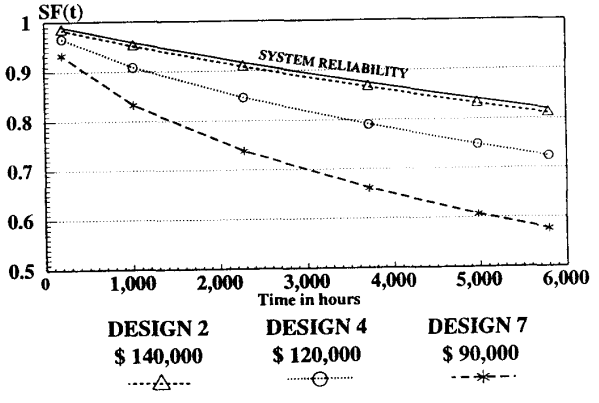


Figure 7. Time-Dependent $\mathcal{SFI}$ for Various Designs

If one of the two mixers fails (assume failure rates described by weif(t/α; β), table 4) the $\mathcal{FOR}$, figure 4, and the operational characteristics of the plant, as shown in figure 5, change appreciably, resulting in a reduction of the $\mathcal{SFI}$ value to 0.6417. For the various operable system states, depicted in figure 6, a corresponding $\mathcal{SFI}$ can be obtained; expressing also the state probabilities as a function of time through the reliability model, the $\mathcal{SFI}$ as a function of time can be obtained, as shown in table 5 and figure 7. As time increases, the system $\mathcal{SFI}$ decreases (no maintenance). The estimation of the $\mathcal{FRI}$ yields 0.8266, for $\mathcal{IC}$ = 6000 hours, implying that the system (on average) without any maintenance action, can withstand 82.66% of both continuous demand variations and equipment failures.

TABLE 5
Effect of C_I^{max} on $\mathcal{SFI}$

Design Case	$\mathcal{SFI}$					
Investment ($)	1	2	3	4	5	6
state	150k	140k	130k	120k	110k	100k
2-2-1	.9946	.9946	.9946	.9946	0.9946	0.9942
1-2-1	.9928	.9759	.8817	.6417	0.3369	0.1224
2-1-1	.9946	.9946	.9946	.9946	0.9925	0.9371
1-1-1	.9928	.9759	.8817	.6417	0.3369	0.1224

In this case the impetus for executing maintenance policies is clear, since there is a substantial deterioration of system performance (from 99.46% to 82.66%). Due to the nature of the batch plant operation in figure 5, preventive maintenance can be effectively considered, especially when equipment is idle.

4. DESIGN OPTIMIZATION

In the analysis so far, decisions regarding the process design itself (*eg*, amount of equipment) were considered fixed. A subsequent question is then, how to obtain a minimum cost (or maximum *s*-expected profit) design, while ensuring that the system can handle process variations and equipment failures. Conceptually, this problem can be posed as the following design optimization of the $\mathcal{FRI}$, subject to a process model, design specifications, and process economics constraints.

$$\max_{\boldsymbol{d}}\{\mathrm{FR}_{\mathcal{IC}}(\boldsymbol{d})\} = \frac{1}{\mathcal{IC}} \cdot \int_{T_a}^{T_b} \sum_{k \in \mathrm{KOP}} P_k(t) \cdot \mathrm{SF}_k(\boldsymbol{d})\, dt. \qquad (5)$$

subject to,

- Process economics ($C_I(\boldsymbol{d}) \leq C_I^{\max}$).

Problem (5) is a parametric-in-C_I^{max} optimization which additionally requires the estimation of $P_k(t)$ & $\mathrm{SF}_k(\boldsymbol{d})$ as a function of the design variables $\boldsymbol{d}$. To overcome this difficulty, an iterative procedure is applied based on a modified Generalized Benders Decomposition coupled with a suitable Gaussian quadrature integration scheme [13] as follows.

By fixing the design variables, the $\mathcal{FRI}$ is estimated through the quadrature formula; this also requires identification

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of the plant $\mathfrak{FOR}$ (for all feasible system states) via the solution of an optimization problem. A master problem is then constructed for updating the design variables, while tradeoff considerations between cost and flexibility-reliability are accounted for by varying the value of $C_I^{\max}$. If a profit term is also considered, this procedure effectively leads to a design featuring a level of flexibility-reliability, that maximize an s-expected profit criterion. The formal design optimization procedure is not presented here, but its application is shown to the multiproduct batch plant design case-study.

The results are summarized in tables 5 & 6 and shown graphically in figures 7 & 8.

TABLE 6
Design Optimization — Summary of Results

Case	C_I	E{Profit}	FR_{3C}	$V_{M\text{-}1,M\text{-}2}$	$V_{R\text{-}1,R\text{-}2}$	$V_{C\text{-}1}$
	150k	242.8k	.8930	1209.2	1813.8	2000.0
	140k		.8898	1041.3	1561.9	2000.0
Optimal	134.8k	252.5k	.8837	968.5	1452.7	1937.0
	130k		.8720	911.3	1367.0	1822.7
	120k	238.3k	.8266	797.5	1196.3	1595.1
	110k	222.8k	.7685	689.9	1034.8	1379.7
	100k		.7164	588.5	882.8	1177.1
	90k		.6240	500.0	742.5	970.6
	80k		.3082	500.0	568.8	758.4
	70k		.1334	324.7	487.1	649.6

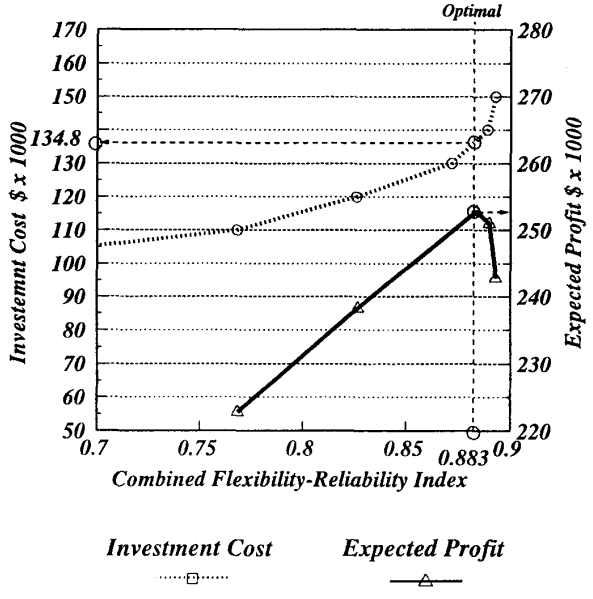


Figure 8. Investment Cost and s-Expected Profit *vs* $\mathfrak{FRI}$

As $C_I^{\max}$ increases in table 6, *ie*, allowing for more investment cost, the volumes of equipment increase and the corresponding $\mathfrak{FRI}$ values increase. As a result, the more expensive designs have an average system performance (in absorbing the demand variations and equipment failures) which is almost identical to the maximum performance that the system can achieve when all the equipment is available (the $\mathfrak{SFI}$). On the other hand, there is a tradeoff, as revealed in figure 8. The design which maximizes s-expected profit corresponds to a less conservative design (smaller volumes, see table 6) with an associated $\mathfrak{FRI}$ of 0.8837; this optimal design requires an investment cost of \$134.8k.

The framework presented here is not, of course, completed to this end; the main objective is to link our framework, which is mainly suitable for process design studies, to existing powerful reliability-maintainability techniques. Such a link will help bridge the gap between chemical engineering practitioners and reliability engineers.

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